Topic 0 | grade C | fails: F2,F3 | n=411

NPMI -0.415 | Jaccard 0.667 | modularity 0.436
Top-10: parking, occupancy, toll, park, lanes, carsharing, san, pricing, parks, atlanta
Journal: TRR 0.75 | peak 2005 | slope -0.057pp/yr | nonfixed 0.07
Rep1: 2013 TRR | Parking in the City
Rep2: 2014 JPT | Commuter Mode Choice and Free Car Parking, Public Transportation Benefits, Showers/Lockers, and Bike Parking a
Rep3: 2008 TRR | Reassessing On-Street Parking

Topic 1 | grade A | fails: none | n=586

NPMI -0.060 | Jaccard 0.667 | modularity 0.070
Top-10: electric, vehicles, electrification, plug, battery, energy, fleet, duty, emissions, buses
Journal: TR-D 0.36 | peak 2025 | slope +0.041pp/yr | nonfixed 0.25
Rep1: 2015 TR-A | Assessing CO 2 emissions of electric vehicles in Germany in 2030
Rep2: 2016 TranspPol | Perspectives of electric mobility: Total cost of ownership of electric vehicles in Germany
Rep3: 2017 TR-D | Impact of road gradient on energy consumption of electric vehicles

Topic 2 | grade A | fails: none | n=406

NPMI -0.126 | Jaccard 0.667 | modularity 0.190
Top-10: horizontal, alignment, curve, vertical, alignments, curves, geometric, sight, mountainous, design
Journal: AAP 0.20 | peak 2009 | slope -0.006pp/yr | nonfixed 0.15
Rep1: 2012 AAP | New geometric design consistency model based on operating speed profiles for road safety evaluation
Rep2: 2016 TRR | Use of Side Friction in Horizontal Curve Design: A Margin of Safety Assessment
Rep3: 2016 TRR | Reliability-Based Design of Horizontal Curves on Two-Lane Rural Highways

Topic 3 | grade A | fails: none | n=677

NPMI -0.173 | Jaccard 0.667 | modularity 0.163
Top-10: pricing, congestion, toll, road, tradable, schemes, tolls, welfare, equilibrium, transport
Journal: TR-A 0.25 | peak 2007 | slope -0.034pp/yr | nonfixed 0.06
Rep1: 2002 JTG | Congestion management and electronic road pricing in Singapore
Rep2: 2009 TranspRev | Equity Effects of Road Pricing: A Review
Rep3: 2014 TranspRev | The Role of Tradable Credit Schemes in Road Traffic Congestion Management

Topic 4 | grade B | fails: F2 | n=965

NPMI -0.226 | Jaccard 0.667 | modularity 0.173
Top-10: mobility, walking, older, adults, commuting, people, home, women, built, social
Journal: JTG 0.26 | peak 2025 | slope +0.054pp/yr | nonfixed 0.25
Rep1: 2021 JTG | To walk or not to walk? Examining non-linear effects of streetscape greenery on walking propensity of older ad
Rep2: 2011 TRR | Transportation, Distance, and Health Care Utilization for Older Adults in Rural and Small Urban Areas
Rep3: 2006 JTG | Accessibility, mobility and transport-related social exclusion

Topic 5 | grade A | fails: none | n=380

NPMI -0.145 | Jaccard 0.818 | modularity 0.148
Top-10: positioning, localization, navigation, gnss, radar, filter, gps, sensor, aided, inertial
Journal: T-ITS 0.66 | peak 2024 | slope +0.029pp/yr | nonfixed 0.22
Rep1: 2020 ITS Mag | Robust Vehicular Localization and Map Matching in Urban Environments Through IMU, GNSS, and Cellular Signals
Rep2: 2009 T-ITS | In-Car Positioning and Navigation Technologies—A Survey
Rep3: 2011 T-ITS | Interacting Multiple Model Filter-Based Sensor Fusion of GPS With In-Vehicle Sensors for Real-Time Vehicle Pos

Topic 6 | grade B | fails: F2 | n=251

NPMI -0.262 | Jaccard 0.667 | modularity 0.216
Top-10: pavement, tire, load, mechanistic, truck, axle, weigh, tires, damage, contact
Journal: TRR 0.95 | peak 2003 | slope -0.042pp/yr | nonfixed 0.03
Rep1: 2006 TRR | Characterization of Truck Traffic in California for Mechanistic-Empirical Design
Rep2: 2007 TRR | Development and Influence of Statewide Axle Load Spectra on Flexible Pavement Performance
Rep3: 2014 TRR | Analytical Approach for Predicting Three-Dimensional Tire–Pavement Contact Load

Topic 7 | grade A | fails: none | n=435

NPMI -0.026 | Jaccard 0.818 | modularity 0.130
Top-10: prediction, forecasting, short, term, neural, traffic, machine, time, seasonal, regression
Journal: TRR 0.43 | peak 2004 | slope -0.027pp/yr | nonfixed 0.09
Rep1: 2015 TR-C | Long short-term memory neural network for traffic speed prediction using remote microwave sensor data
Rep2: 2014 TR-C | Short-term traffic forecasting: Where we are and where we’re going
Rep3: 2007 TRR | Adaptive Seasonal Time Series Models for Forecasting Short-Term Traffic Flow

Topic 8 | grade A | fails: none | n=917

NPMI -0.022 | Jaccard 0.667 | modularity 0.137
Top-10: accessibility, transit, equity, spatial, access, oriented, public, measuring, minute, transport
Journal: JTG 0.32 | peak 2025 | slope +0.058pp/yr | nonfixed 0.17
Rep1: 2012 JTG | Measuring accessibility: positive and normative implementations of various accessibility indicators
Rep2: 2013 JTG | Towards equitable transit: examining transit accessibility and social need in Toronto, Canada, 1996–2006
Rep3: 2017 JTG | Assessing equity in transport accessibility to work and study: The Bogotá region

Topic 9 | grade B | fails: F2 | n=475

NPMI -0.205 | Jaccard 0.667 | modularity 0.079
Top-10: data, big, analytics, media, incident, transportation, crowdsourced, clustering, reports, mining
Journal: TRR 0.25 | peak 2020 | slope +0.027pp/yr | nonfixed 0.15
Rep1: 2018 T-ITS | Big Data Analytics in Intelligent Transportation Systems: A Survey
Rep2: 2016 TR-C | From Twitter to detector: Real-time traffic incident detection using social media data
Rep3: 2018 TR-C | Recent applications of big data analytics in railway transportation systems: A survey

Topic 10 | grade A | fails: none | n=448

NPMI -0.068 | Jaccard 0.818 | modularity 0.095
Top-10: simulation, microsimulation, microscopic, traffic, calibration, models, vissim, freeway, validation, toll
Journal: TRR 0.75 | peak 2000 | slope -0.109pp/yr | nonfixed 0.06
Rep1: 2005 TRR | Analysis and Comparison of Microscopic Traffic Flow Models with Real Traffic Microscopic Data
Rep2: 2005 TRR | Hybrid Mesoscopic-Microscopic Traffic Simulation
Rep3: 2005 TRR | Development and Evaluation of a Procedure for the Calibration of Simulation Models

Topic 11 | grade A | fails: none | n=664

NPMI 0.028 | Jaccard 0.818 | modularity 0.139
Top-10: crash, severity, random, injury, crashes, bayesian, heterogeneity, multivariate, parameters, severities
Journal: AAP 0.59 | peak 2019 | slope +0.006pp/yr | nonfixed 0.24
Rep1: 2006 TRR | Bayesian Multivariate Poisson Regression for Models of Injury Count, by Severity
Rep2: 2007 TRR | Multivariate Poisson-Lognormal Models for Jointly Modeling Crash Frequency by Severity
Rep3: 2018 AAP | Factors affecting motorcyclists’ injury severities: An empirical assessment using random parameters logit mode

Topic 12 | grade A | fails: none | n=326

NPMI -0.019 | Jaccard 0.818 | modularity 0.119
Top-10: power, dc, energy, control, aircraft, cell, fuel, storage, hybrid, electric
Journal: T-TE 0.76 | peak 2023 | slope +0.034pp/yr | nonfixed 0.83
Rep1: 2016 T-TE | All-Electric Ship Design: From Electrical Propulsion to Integrated Electrical and Electronic Power Systems
Rep2: 2022 T-TE | All-Electric NASA N3-X Aircraft Electric Power Systems
Rep3: 2018 T-TE | Investigation on the Selection of Electric Power System Architecture for Future More Electric Aircraft

Topic 13 | grade A | fails: none | n=573

NPMI -0.011 | Jaccard 0.818 | modularity 0.149
Top-10: traffic, flow, fundamental, estimation, link, macroscopic, diagram, origin, networks, destination
Journal: TR-C 0.29 | peak 2014 | slope -0.009pp/yr | nonfixed 0.04
Rep1: 2014 T-ITS | Traffic Flow Prediction for Road Transportation Networks With Limited Traffic Data
Rep2: 2007 TRR | Offline Calibration of Dynamic Traffic Assignment
Rep3: 2003 TRR | Toward Benchmarking of Microscopic Traffic Flow Models

Topic 14 | grade A | fails: none | n=292

NPMI -0.082 | Jaccard 0.667 | modularity 0.227
Top-10: modulus, resilient, pavement, subgrade, base, deflectometer, falling, unbound, stiffness, materials
Journal: TRR 0.96 | peak 2005 | slope -0.057pp/yr | nonfixed 0.03
Rep1: 2013 TRR | Evaluation of Permanent Deformation Characteristics of Unbound Granular Materials by Means of Multistage Repea
Rep2: 2004 TRR | Correlation Between Resilient Modulus, Moisture Variation, and Soil Suction for Subgrade Soils
Rep3: 2005 TRR | Variation of Resilient Modulus with Soil Suction for Compacted Subgrade Soils

Topic 15 | grade B | fails: F2 | n=435

NPMI -0.197 | Jaccard 0.667 | modularity 0.116
Top-10: detection, object, feature, tracking, recognition, attention, identification, pedestrian, scenes, counting
Journal: T-ITS 0.77 | peak 2023 | slope +0.042pp/yr | nonfixed 0.19
Rep1: 2022 T-ITS | Edge YOLO: Real-Time Intelligent Object Detection System Based on Edge-Cloud Cooperation in Autonomous Vehicle
Rep2: 2019 ETRR | Vision-based vehicle detection and counting system using deep learning in highway scenes
Rep3: 2021 T-ITS | A Review and Comparative Study on Probabilistic Object Detection in Autonomous Driving

Topic 16 | grade A | fails: none | n=629

NPMI -0.038 | Jaccard 0.667 | modularity 0.083
Top-10: bus, transit, scheduling, passenger, stop, demand, disruptions, service, timetable, responsive
Journal: TRR 0.26 | peak 2020 | slope +0.028pp/yr | nonfixed 0.09
Rep1: 2022 TR-C | A survey on demand-responsive public bus systems
Rep2: 2015 TransSci | Optimizing Bus Bridging Services in Response to Disruptions of Urban Transit Rail Networks
Rep3: 2021 TR-B | On the design of an optimal flexible bus dispatching system with modular bus units: Using the three-dimensiona

Topic 17 | grade A | fails: none | n=447

NPMI -0.006 | Jaccard 1.000 | modularity 0.082
Top-10: autonomous, testing, scenario, vehicles, automated, safety, scenarios, driving, generation, risk
Journal: T-ITS 0.27 | peak 2025 | slope +0.045pp/yr | nonfixed 0.34
Rep1: 2021 AAP | A review of surrogate safety measures and their applications in connected and automated vehicles safety modeli
Rep2: 2021 TRR | Corner Case Generation and Analysis for Safety Assessment of Autonomous Vehicles
Rep3: 2016 T-ITS | Accelerated Evaluation of Automated Vehicles Safety in Lane-Change Scenarios Based on Importance Sampling Tech

Topic 19 | grade A | fails: none | n=563

NPMI -0.176 | Jaccard 0.818 | modularity 0.143
Top-10: emissions, carbon, transport, co2, climate, emission, environmental, air, policy, greenhouse
Journal: TR-D 0.35 | peak 2010 | slope +0.011pp/yr | nonfixed 0.15
Rep1: 2007 JTG | Transport and climate change: a review
Rep2: 2017 TranspPol | Road transport and CO 2 emissions: What are the challenges?
Rep3: 2021 TR-D | Transport infrastructure and CO2 emissions in the OECD over the long run

Topic 20 | grade A | fails: none | n=422

NPMI -0.126 | Jaccard 0.818 | modularity 0.174
Top-10: privacy, secure, security, authentication, preserving, scheme, vehicular, protocol, vanets, intrusion
Journal: T-ITS 0.77 | peak 2022 | slope +0.043pp/yr | nonfixed 0.13
Rep1: 2021 T-ITS | Efficient and Secure Routing Protocol Based on Artificial Intelligence Algorithms With UAV-Assisted for Vehicu
Rep2: 2015 T-ITS | ART: An Attack-Resistant Trust Management Scheme for Securing Vehicular Ad Hoc Networks
Rep3: 2017 T-ITS | EAAP: Efficient Anonymous Authentication With Conditional Privacy-Preserving Scheme for Vehicular Ad Hoc Netwo

Topic 21 | grade A | fails: none | n=523

NPMI -0.113 | Jaccard 0.818 | modularity 0.108
Top-10: acceptance, autonomous, automated, trust, vehicles, intentions, intention, adoption, use, willingness
Journal: TR-F 0.33 | peak 2020 | slope +0.048pp/yr | nonfixed 0.22
Rep1: 2021 TR-F | A structural equation modeling approach for the acceptance of driverless automated shuttles based on construct
Rep2: 2022 TR-A | Trust and perceived risk: How different manifestations affect the adoption of autonomous vehicles
Rep3: 2022 TRR | Modeling Users' Adoption of Shared Autonomous Vehicles Employing Actual Ridership Experiences

Topic 22 | grade A | fails: none | n=951

NPMI 0.015 | Jaccard 0.818 | modularity 0.099
Top-10: data, travel, card, gps, trip, patterns, destination, smart, activity, origin
Journal: T-ITS 0.19 | peak 2020 | slope +0.060pp/yr | nonfixed 0.14
Rep1: 2018 JTG | Spatial variations in urban public ridership derived from GPS trajectories and smart card data
Rep2: 2016 JTG | Understanding commuting patterns using transit smart card data
Rep3: 2001 TRR | Elimination of the Travel Diary: Experiment to Derive Trip Purpose from Global Positioning System Travel Data

Topic 23 | grade A | fails: none | n=662

NPMI 0.026 | Jaccard 0.818 | modularity 0.100
Top-10: rail, china, high, speed, railway, economic, evidence, growth, transport, railways
Journal: JTG 0.23 | peak 2002 | slope -0.006pp/yr | nonfixed 0.11
Rep1: 2017 JTG | Impact of high-speed rail on regional economic disparity in China
Rep2: 2017 JTG | Impacts of high-speed rail lines on the city network in China
Rep3: 2014 JTG | Impacts of high speed rail on railroad network accessibility in China

Topic 24 | grade A | fails: none | n=166

NPMI -0.109 | Jaccard 0.538 | modularity 0.176
Top-10: management, information, maintenance, geographic, highway, implementation, development, asset, department, wisconsin
Journal: TRR 0.87 | peak 2000 | slope -0.082pp/yr | nonfixed 0.01
Rep1: 2000 TRR | Integrating Genetic Algorithms and Geographic Information System to Optimize Highway Alignments
Rep2: 2000 CACIE | Preliminary Highway Design with Genetic Algorithms and Geographic Information Systems
Rep3: 2001 TRR | Method of Improving Pedestrian Safety Proactively with Geographic Information Systems: Example from a College

Topic 25 | grade B | fails: F2 | n=275

NPMI -0.275 | Jaccard 0.667 | modularity 0.199
Top-10: bridge, barrier, guardrail, beam, track, testing, midwest, concrete, test, steel
Journal: TRR 0.97 | peak 2000 | slope -0.045pp/yr | nonfixed 0.01
Rep1: 2001 TRR | Improvements to the Weak-Post W-Beam Guardrail
Rep2: 2001 TRR | Three-Strand Cable Median Barrier in North Carolina: In-Service Evaluation
Rep3: 2007 TRR | Performance of Steel-Post, W-Beam Guardrail Systems

Topic 26 | grade A | fails: none | n=273

NPMI -0.102 | Jaccard 0.818 | modularity 0.120
Top-10: concrete, pavement, pavements, overlay, overlays, asphalt, performance, reflective, rehabilitation, cracking
Journal: TRR 0.98 | peak 2000 | slope -0.074pp/yr | nonfixed 0.02
Rep1: 2007 TRR | Guidelines for Using Geosynthetics with Hot-Mix Asphalt Overlays to Reduce Reflective Cracking
Rep2: 2009 TRR | Pervious Concrete Pavement
Rep3: 2011 TRR | Use of Data from Specific Pavement Studies Experiment 5 in the Long-Term Pavement Performance Program to Compa

Topic 27 | grade B | fails: F2 | n=362

NPMI -0.470 | Jaccard 0.538 | modularity 0.224
Top-10: freight, demand, price, market, forecasting, elasticities, shipment, industry, airline, commodity
Journal: TR-E 0.19 | peak 2012 | slope -0.001pp/yr | nonfixed 0.09
Rep1: 2009 TR-E | Air transport demand and economic growth in Brazil: A time series analysis
Rep2: 2009 TRR | Freight Generation Models
Rep3: 2019 TRR | Comparative Assessment of Industrial Classification Systems for Modeling Freight Production and Freight Trip P

Topic 28 | grade C | fails: F1,F2 | n=185

NPMI -0.389 | Jaccard 0.429 | modularity 0.245
Top-10: aerial, unmanned, radar, detection, inspection, monitoring, measurement, ranging, millimeter, sensing
Journal: TRR 0.32 | peak 2018 | slope +0.003pp/yr | nonfixed 0.11
Rep1: 2020 TR-A | Applications of unmanned aerial vehicle (UAV) in road safety, traffic and highway infrastructure management: R
Rep2: 2021 TRR | Revealing Safety Risks of Unmanned Aerial Vehicles in Construction
Rep3: 2017 T-ITS | Video Processing From Electro-Optical Sensors for Object Detection and Tracking in a Maritime Environment: A S

Topic 29 | grade A | fails: none | n=392

NPMI -0.002 | Jaccard 0.818 | modularity 0.162
Top-10: 3d, object, point, lidar, detection, semantic, depth, monocular, cloud, segmentation
Journal: T-ITS 0.67 | peak 2025 | slope +0.044pp/yr | nonfixed 0.29
Rep1: 2020 T-ITS | Deep Multi-Modal Object Detection and Semantic Segmentation for Autonomous Driving: Datasets, Methods, and Cha
Rep2: 2021 T-ITS | Deep Learning for Image and Point Cloud Fusion in Autonomous Driving: A Review
Rep3: 2022 T-ITS | Comprehensive Review of Deep Learning-Based 3D Point Cloud Completion Processing and Analysis

Topic 30 | grade A | fails: none | n=693

NPMI -0.037 | Jaccard 0.818 | modularity 0.089
Top-10: injury, crashes, crash, injuries, risk, accidents, severity, factors, fatal, motor
Journal: AAP 0.76 | peak 2002 | slope -0.079pp/yr | nonfixed 0.05
Rep1: 2012 AAP | Driver-injury severity in single-vehicle crashes in California: A mixed logit analysis of heterogeneity due to
Rep2: 2006 AAP | Bicyclist injury severities in bicycle–motor vehicle accidents
Rep3: 2015 AMAR | The temporal stability of factors affecting driver-injury severities in single-vehicle crashes: Some empirical

Topic 31 | grade A | fails: none | n=534

NPMI -0.013 | Jaccard 0.667 | modularity 0.078
Top-10: asphalt, mixtures, fatigue, damage, binders, cracking, binder, aging, properties, mixture
Journal: TRR 0.88 | peak 2002 | slope -0.037pp/yr | nonfixed 0.07
Rep1: 2014 TRR | Development of a Failure Criterion for Asphalt Mixtures under Different Modes of Fatigue Loading
Rep2: 2002 TRR | Surface Energy Measurement of Asphalt and Its Application to Predicting Fatigue and Healing in Asphalt Mixture
Rep3: 2012 TRR | Characterization of Fracture Properties of Asphalt Mixtures as Measured by Semicircular Bend Test and Indirect

Topic 32 | grade B | fails: F2 | n=260

NPMI -0.330 | Jaccard 0.538 | modularity 0.230
Top-10: bridge, bridges, load, rating, steel, inspection, deterioration, element, maintenance, reliability
Journal: TRR 0.86 | peak 2000 | slope -0.047pp/yr | nonfixed 0.05
Rep1: 2022 TRR | Bridge Deck Deterioration: Reasons and Patterns
Rep2: 2004 TRR | Bridge Deck Management System with Integrated Life-Cycle Cost Optimization
Rep3: 2001 TRR | Highway Bridge Inspection: State-of-the-Practice Survey

Topic 33 | grade A | fails: none | n=719

NPMI -0.152 | Jaccard 0.818 | modularity 0.191
Top-10: ride, sharing, demand, ridesharing, services, matching, bike, sourcing, pooling, hailing
Journal: TR-C 0.20 | peak 2021 | slope +0.063pp/yr | nonfixed 0.08
Rep1: 2017 TransSci | Stable Matching for Dynamic Ride-Sharing Systems
Rep2: 2013 ITS Mag | Relocation Strategies and Algorithms for Free-Floating Car Sharing Systems
Rep3: 2015 TR-B | The benefits of meeting points in ride-sharing systems

Topic 34 | grade A | fails: none | n=568

NPMI -0.030 | Jaccard 0.818 | modularity 0.143
Top-10: carbon, emissions, emission, environmental, co2, transportation, reduction, freight, dioxide, sector
Journal: TR-D 0.38 | peak 2015 | slope +0.020pp/yr | nonfixed 0.19
Rep1: 2009 TRR | Trade-Off between Carbon Dioxide Emissions and Logistics Costs Based on Multiobjective Optimization
Rep2: 2019 TR-D | Relationship between the development and CO2 emissions of transport sector in China
Rep3: 2022 JTTE | Road life-cycle carbon dioxide emissions and emission reduction technologies: A review

Topic 35 | grade A | fails: none | n=631

NPMI -0.091 | Jaccard 0.667 | modularity 0.165
Top-10: safety, drivers, phone, perceptions, road, cyclists, use, parents, attitudes, perception
Journal: AAP 0.35 | peak 2009 | slope +0.007pp/yr | nonfixed 0.12
Rep1: 2022 TR-F | Identifying the determinants and understanding their effect on the perception of safety, security, and comfort
Rep2: 2017 TR-F | Mobile phone use while driving-literary review
Rep3: 2019 AAP | A dual perspective on risk perception and its effect on safety behavior: A moderated mediation model of safety

Topic 36 | grade A | fails: none | n=422

NPMI -0.138 | Jaccard 0.818 | modularity 0.262
Top-10: shopping, online, delivery, commerce, platform, store, food, consumer, grocery, brand
Journal: TR-E 0.40 | peak 2023 | slope +0.029pp/yr | nonfixed 0.14
Rep1: 2022 TranspPol | Analysis and modeling of changes in online shopping behavior due to Covid-19 pandemic: A Florida case study
Rep2: 2022 TRIP | Investigation of online grocery shopping and delivery preference before, during, and after COVID-19
Rep3: 2018 TRR | Accessibility or Innovation? Store Shopping Trips versus Online Shopping

Topic 37 | grade A | fails: none | n=553

NPMI -0.044 | Jaccard 1.000 | modularity 0.080
Top-10: control, platoon, cooperative, connected, platoons, distributed, platooning, cruise, communication, adaptive
Journal: T-ITS 0.51 | peak 2023 | slope +0.055pp/yr | nonfixed 0.21
Rep1: 2015 TRR | Cooperative Adaptive Cruise Control
Rep2: 2018 TR-B | Cooperative platoon control for a mixed traffic flow including human drive vehicles and connected and autonomo
Rep3: 2019 TRR | Cooperative Adaptive Cruise Control for a Platoon of Connected and Autonomous Vehicles considering Dynamic Inf

Topic 38 | grade A | fails: none | n=392

NPMI -0.075 | Jaccard 0.818 | modularity 0.194
Top-10: 5g, radio, mobile, wireless, 6g, communications, networks, spectrum, channel, challenges
Journal: VT Mag 0.67 | peak 2019 | slope +0.021pp/yr | nonfixed 0.71
Rep1: 2019 VT Mag | 6G Wireless Networks: Vision, Requirements, Architecture, and Key Technologies
Rep2: 2016 VT Mag | Massive MIMO and mmWave for 5G Wireless HetNet: Potential Benefits and Challenges
Rep3: 2020 VT Mag | Machine Learning for 6G Wireless Networks: Carrying Forward Enhanced Bandwidth, Massive Access, and Ultrarelia

Topic 39 | grade A | fails: none | n=402

NPMI 0.053 | Jaccard 0.818 | modularity 0.042
Top-10: energy, electric, hybrid, management, vehicles, strategy, plug, eco, fuel, cell
Journal: T-TE 0.40 | peak 2024 | slope +0.040pp/yr | nonfixed 0.61
Rep1: 2016 TRR | Data-Driven Reinforcement Learning–Based Real-Time Energy Management System for Plug-In Hybrid Electric Vehicl
Rep2: 2021 T-TE | Double Deep Reinforcement Learning-Based Energy Management for a Parallel Hybrid Electric Vehicle With Engine
Rep3: 2022 T-TE | Visual Detection and Deep Reinforcement Learning-Based Car Following and Energy Management for Hybrid Electric

Topic 40 | grade B | fails: F2 | n=506

NPMI -0.314 | Jaccard 0.667 | modularity 0.263
Top-10: signs, zones, work, flashing, message, red, yellow, evaluation, intersections, rumble
Journal: TRR 0.88 | peak 2006 | slope -0.076pp/yr | nonfixed 0.04
Rep1: 2014 TRR | Driver Yielding at Traffic Control Signals, Pedestrian Hybrid Beacons, and Rectangular Rapid-Flashing Beacons
Rep2: 2007 TRR | Development and Field Evaluation of Variable Advisory Speed Limit System for Work Zones
Rep3: 2004 TRR | Safety Evaluation of Centerline Rumble Strips: Crash and Driver Behavior Analysis

Topic 41 | grade A | fails: none | n=548

NPMI -0.189 | Jaccard 0.818 | modularity 0.123
Top-10: container, shipping, liner, port, ship, terminals, vessel, inland, repositioning, containers
Journal: TR-E 0.29 | peak 2015 | slope +0.006pp/yr | nonfixed 0.08
Rep1: 2008 TransSci | Ship Scheduling and Network Design for Cargo Routing in Liner Shipping
Rep2: 2008 JTG | The effect of high fuel costs on liner service configuration in container shipping
Rep3: 2005 TR-E | The container shipping network design problem with empty container repositioning

Topic 42 | grade A | fails: none | n=689

NPMI -0.089 | Jaccard 0.818 | modularity 0.105
Top-10: crash, risk, severity, prediction, crashes, machine, accident, data, using, mining
Journal: AAP 0.45 | peak 2025 | slope +0.042pp/yr | nonfixed 0.14
Rep1: 2017 AAP | Comparison of four statistical and machine learning methods for crash severity prediction
Rep2: 2019 TRR | Real-Time Crash Risk Prediction using Long Short-Term Memory Recurrent Neural Network
Rep3: 2019 TRR | Improved Support Vector Machine Models for Work Zone Crash Injury Severity Prediction and Analysis

Topic 43 | grade B | fails: F2 | n=324

NPMI -0.246 | Jaccard 0.667 | modularity 0.200
Top-10: aggregate, asphalt, friction, pavements, porous, pavement, graded, aggregates, coarse, chip
Journal: TRR 0.93 | peak 2000 | slope -0.052pp/yr | nonfixed 0.06
Rep1: 2015 TRR | Comparison of Permeability and Clogging Characteristics of Porous Asphalt and Pervious Concrete Pavement Mater
Rep2: 2001 TRR | Correlation of Fine Aggregate Imaging Shape Indices with Asphalt Mixture Performance
Rep3: 2003 TRR | Aggregate Imaging System for Characterizing the Shape of Fine and Coarse Aggregates

Topic 44 | grade A | fails: none | n=478

NPMI -0.032 | Jaccard 0.818 | modularity 0.038
Top-10: electric, charging, adoption, vehicle, plug, battery, ev, preferences, vehicles, infrastructure
Journal: TR-D 0.30 | peak 2025 | slope +0.045pp/yr | nonfixed 0.21
Rep1: 2018 TR-D | A review of consumer preferences of and interactions with electric vehicle charging infrastructure
Rep2: 2016 TR-D | Incentives for promoting Battery Electric Vehicle (BEV) adoption in Norway
Rep3: 2013 TRR | Locating Electric Vehicle Charging Stations

Topic 45 | grade A | fails: none | n=521

NPMI -0.011 | Jaccard 0.818 | modularity 0.139
Top-10: phase, current, control, drives, fault, pmsm, torque, predictive, motor, observer
Journal: T-TE 0.98 | peak 2025 | slope +0.058pp/yr | nonfixed 0.99
Rep1: 2021 T-TE | Speed Sensorless Model Predictive Current Control Based on Finite Position Set for PMSHM Drives
Rep2: 2022 T-TE | An Improved Implicit Model Predictive Current Control With Continuous Control Set for PMSM Drives
Rep3: 2020 T-TE | Model Predictive Control of Switched Reluctance Starter/Generator With Torque Sharing and Compensation

Topic 46 | grade A | fails: none | n=482

NPMI -0.023 | Jaccard 0.818 | modularity 0.110
Top-10: travel, time, reliability, origin, estimation, destination, probe, times, freeway, data
Journal: TRR 0.69 | peak 2005 | slope -0.052pp/yr | nonfixed 0.05
Rep1: 2010 TRR | Travel Time Forecasting and Dynamic Origin-Destination Estimation for Freeways Based on Bluetooth Traffic Moni
Rep2: 2011 TRR | Analytic Relationships between Travel Time Reliability Measures
Rep3: 2005 TRR | Monitoring and Predicting Freeway Travel Time Reliability

Topic 47 | grade A | fails: none | n=805

NPMI -0.085 | Jaccard 0.818 | modularity 0.124
Top-10: choice, mode, transit, public, ridership, travel, preference, rail, transport, preferences
Journal: TRR 0.30 | peak 2021 | slope +0.036pp/yr | nonfixed 0.20
Rep1: 2010 JTG | Determinants of transport mode choice: a comparison of Germany and the USA
Rep2: 2020 TRR | Role of Attitudes in Transit and Auto Users' Mode Choice of Ridesourcing
Rep3: 2016 TR-A | Factors affecting car ownership and mode choice in rail transit-supported suburbs of a large Chinese city

Topic 48 | grade A | fails: none | n=620

NPMI -0.184 | Jaccard 0.818 | modularity 0.120
Top-10: bike, sharing, bicycle, bikes, dockless, share, bikeshare, usage, scooter, bikesharing
Journal: TRR 0.18 | peak 2020 | slope +0.048pp/yr | nonfixed 0.22
Rep1: 2019 JTG | Spatiotemporal comparative analysis of scooter-share and bike-share usage patterns in Washington, D.C.
Rep2: 2018 IJST | Understanding the usage of dockless bike sharing in Singapore
Rep3: 2020 TR-D | Built environment effects on the integration of dockless bike-sharing and the metro

Topic 49 | grade C | fails: F2,F3 | n=389

NPMI -0.836 | Jaccard 0.538 | modularity 0.759
Top-10: page, papers, journal, lab, conference, fundamental, tgrg, selected, retrieval, synthesis
Journal: CACIE 0.15 | peak 2003 | slope -0.013pp/yr | nonfixed 0.11
Rep1: 2020 TR-B | Fifty years of the bottleneck model: A bibliometric review and future research directions
Rep2: 2011 TR-B | Clockwise hysteresis loops in the Macroscopic Fundamental Diagram: An effect of network instability
Rep3: 2005 AAP | Experimental evaluation of hotspot identification methods

Topic 50 | grade A | fails: none | n=882

NPMI -0.040 | Jaccard 0.818 | modularity 0.029
Top-10: charging, electric, stations, battery, scheduling, station, ev, optimal, vehicle, energy
Journal: T-ITS 0.22 | peak 2024 | slope +0.090pp/yr | nonfixed 0.28
Rep1: 2011 TRR | Optimal Location of Charging Stations for Electric Vehicles in a Neighborhood in Lisbon, Portugal
Rep2: 2012 TR-B | Optimal deployment of public charging stations for plug-in hybrid electric vehicles
Rep3: 2018 TRR | Planning of Fast-Charging Stations for a Battery Electric Bus System under Energy Consumption Uncertainty

Topic 51 | grade A | fails: none | n=598

NPMI -0.138 | Jaccard 0.818 | modularity 0.173
Top-10: driving, driver, following, human, steering, assistance, collision, autonomous, car, avoidance
Journal: T-ITS 0.33 | peak 2018 | slope +0.014pp/yr | nonfixed 0.23
Rep1: 2017 T-ITS | Driving Style Recognition for Intelligent Vehicle Control and Advanced Driver Assistance: A Survey
Rep2: 2019 TRR | Influence of Autonomous Vehicles on Car-Following Behavior of Human Drivers
Rep3: 2021 TR-C | Toward human-vehicle collaboration: Review and perspectives on human-centered collaborative automated driving

Topic 52 | grade A | fails: none | n=948

NPMI 0.001 | Jaccard 0.818 | modularity 0.101
Top-10: crashes, crash, severity, injury, safety, factors, rural, pedestrian, random, fatal
Journal: TRR 0.53 | peak 2017 | slope +0.040pp/yr | nonfixed 0.16
Rep1: 2020 IJTST | Factors affecting injury severity in vehicle-pedestrian crashes: A day-of-week analysis using random parameter
Rep2: 2014 AAP | Mixed logit model-based driver injury severity investigations in single- and multi-vehicle crashes on rural tw
Rep3: 2019 TRR | Modeling Pedestrian Injury Severity in Pedestrian-Vehicle Crashes in Rural and Urban Areas: Mixed Logit Model

Topic 53 | grade A | fails: none | n=415

NPMI -0.122 | Jaccard 0.818 | modularity 0.183
Top-10: vehicular, communications, communication, networks, v2v, channel, hoc, ad, v2x, wireless
Journal: T-ITS 0.62 | peak 2015 | slope +0.019pp/yr | nonfixed 0.25
Rep1: 2018 T-ITS | New Multi-Hop Clustering Algorithm for Vehicular Ad Hoc Networks
Rep2: 2017 VT Mag | LTE-V for Sidelink 5G V2X Vehicular Communications: A New 5G Technology for Short-Range Vehicle-to-Everything
Rep3: 2004 T-ITS | Challenges of Intervehicle Ad Hoc Networks

Topic 54 | grade A | fails: none | n=232

NPMI -0.163 | Jaccard 0.818 | modularity 0.225
Top-10: lithium, ion, batteries, battery, thermal, solid, degradation, cell, membrane, cells
Journal: eTransp 0.58 | peak 2025 | slope +0.026pp/yr | nonfixed 0.97
Rep1: 2021 T-TE | A Comprehensive Review on the Characteristics and Modeling of Lithium-Ion Battery Aging
Rep2: 2022 eTransp | Critical review of life cycle assessment of lithium-ion batteries for electric vehicles: A lifespan perspectiv
Rep3: 2021 eTransp | Comprehensively analysis the failure evolution and safety evaluation of automotive lithium ion battery

Topic 55 | grade A | fails: none | n=346

NPMI -0.168 | Jaccard 0.667 | modularity 0.232
Top-10: traction, railway, power, train, track, electrical, railways, current, rail, electrified
Journal: T-TE 0.53 | peak 2022 | slope +0.030pp/yr | nonfixed 0.76
Rep1: 2022 T-TE | Components of Electrical Power Systems in More and All-Electric Aircraft: A Review
Rep2: 2022 RailEng | Pantograph–catenary electrical contact system of high-speed railways: recent progress, challenges, and outlook
Rep3: 2018 TRR | Full-Scale Model Testing on Ballasted High-Speed Railway: Dynamic Responses and Accumulated Settlements

Topic 56 | grade B | fails: F2 | n=535

NPMI -0.221 | Jaccard 0.667 | modularity 0.138
Top-10: loop, detection, vehicle, detector, sensors, video, data, detectors, inductive, monitoring
Journal: TRR 0.54 | peak 2005 | slope -0.057pp/yr | nonfixed 0.13
Rep1: 2005 TRR | Traffic Measurement and Vehicle Classification with Single Magnetic Sensor
Rep2: 2001 TRR | Freeway Performance Measurement System: Mining Loop Detector Data
Rep3: 2010 TRR | Data Collection of Freeway Travel Time Ground Truth with Bluetooth Sensors

Topic 57 | grade B | fails: F2 | n=386

NPMI -0.417 | Jaccard 0.538 | modularity 0.134
Top-10: vehicles, automated, autonomous, automation, ethical, connected, trust, electronics, retracted, responsibility
Journal: AAP 0.13 | peak 2020 | slope +0.022pp/yr | nonfixed 0.26
Rep1: 2017 TRR | Economic Effects of Automated Vehicles
Rep2: 2015 TR-A | Preparing a nation for autonomous vehicles: opportunities, barriers and policy recommendations
Rep3: 2020 TR-F | Attitudes towards privately-owned and shared autonomous vehicles

Topic 58 | grade A | fails: none | n=221

NPMI -0.085 | Jaccard 0.667 | modularity 0.091
Top-10: blockchain, secure, enabled, privacy, authentication, preserving, internet, vehicular, supply, scheme
Journal: T-ITS 0.57 | peak 2022 | slope +0.025pp/yr | nonfixed 0.14
Rep1: 2018 T-ITS | CreditCoin: A Privacy-Preserving Blockchain-Based Incentive Announcement Network for Communications of Smart V
Rep2: 2020 TR-E | Blockchain technology in supply chain operations: Applications, challenges and research opportunities
Rep3: 2019 TR-E | Blockchain-technology-supported platforms for diamond authentication and certification in luxury supply chains

Topic 59 | grade B | fails: F2 | n=608

NPMI -0.222 | Jaccard 0.667 | modularity 0.103
Top-10: transport, sustainable, transportation, sustainability, research, geography, policy, urban, cities, countries
Journal: TRR 0.19 | peak 2005 | slope -0.007pp/yr | nonfixed 0.24
Rep1: 2017 TranspRev | Cycling towards a more sustainable transport future
Rep2: 2018 TranspPol | Towards sustainable urban transport in Singapore: Policy instruments and mobility trends
Rep3: 2017 TranspPol | Public transport users' and policy makers' perceptions of integrated public transport systems

Topic 60 | grade B | fails: F2 | n=290

NPMI -0.302 | Jaccard 0.667 | modularity 0.328
Top-10: stabilization, soils, soil, water, runoff, lime, ash, salt, chloride, fly
Journal: TRR 0.94 | peak 2000 | slope -0.068pp/yr | nonfixed 0.01
Rep1: 2001 TRR | Mechanisms of Soil Stabilization with Liquid Ionic Stabilizer
Rep2: 2006 TRR | Recommendations for Stabilization of High-Sulfate Soils in Texas
Rep3: 2000 TRR | Field and Laboratory Evaluation of Cement Kiln Dust as a Soil Stabilizer

Topic 61 | grade A | fails: none | n=924

NPMI -0.092 | Jaccard 0.818 | modularity 0.168
Top-10: lane, driving, behavior, changing, driver, drivers, speed, intersections, simulator, overtaking
Journal: TRR 0.30 | peak 2018 | slope +0.029pp/yr | nonfixed 0.13
Rep1: 2007 TRR | General Lane-Changing Model MOBIL for Car-Following Models
Rep2: 2014 IJTST | Analysis of Discretionary Lane Changing Parameters on Freeways
Rep3: 2021 AAP | A proactive lane-changing risk prediction framework considering driving intention recognition and different la

Topic 62 | grade A | fails: none | n=595

NPMI -0.057 | Jaccard 0.818 | modularity 0.103
Top-10: train, railway, optimization, rescheduling, timetable, timetabling, rail, rolling, passenger, trains
Journal: T-ITS 0.20 | peak 2016 | slope +0.038pp/yr | nonfixed 0.11
Rep1: 2016 TR-B | Energy-efficient metro train rescheduling with uncertain time-variant passenger demands: An approximate dynami
Rep2: 2022 TR-B | Joint optimization of train timetabling and rolling stock circulation planning: A novel flexible train composi
Rep3: 2018 TR-B | Passenger demand oriented train scheduling and rolling stock circulation planning for an urban rail transit li

Topic 63 | grade B | fails: F2 | n=266

NPMI -0.297 | Jaccard 0.667 | modularity 0.115
Top-10: alcohol, driving, impaired, drinking, drug, drink, influence, prevalence, impairment, related
Journal: AAP 0.74 | peak 2000 | slope -0.025pp/yr | nonfixed 0.02
Rep1: 2012 AAP | The effects of cannabis and alcohol on simulated driving: Influences of dose and experience
Rep2: 2017 AAP | Effects of acute alcohol consumption on measures of simulated driving: A systematic review and meta-analysis
Rep3: 2010 AAP | The effects of cannabis and alcohol on simulated arterial driving: Influences of driving experience and task d

Topic 64 | grade A | fails: none | n=330

NPMI 0.009 | Jaccard 0.818 | modularity 0.117
Top-10: riders, motorcycle, drivers, motorcyclists, riding, risky, china, behaviors, factors, vietnam
Journal: AAP 0.32 | peak 2012 | slope +0.002pp/yr | nonfixed 0.16
Rep1: 2013 AAP | Risk factors associated with traffic violations and accident severity in China
Rep2: 2019 TRR | Crash Involvement and Risky Riding Behaviors among Delivery Riders in China: The Role of Working Conditions
Rep3: 2012 TRR | Traffic Safety for Electric Bike Riders in China

Topic 65 | grade B | fails: F2 | n=258

NPMI -0.237 | Jaccard 0.667 | modularity 0.148
Top-10: aircraft, flight, trajectory, prediction, ship, learning, ais, airspace, vessel, maritime
Journal: T-ITS 0.34 | peak 2025 | slope +0.021pp/yr | nonfixed 0.08
Rep1: 2022 T-ITS | Vessel Trajectory Prediction in Maritime Transportation: Current Approaches and Beyond
Rep2: 2012 T-ITS | Maritime Traffic Monitoring Based on Vessel Detection, Tracking, State Estimation, and Trajectory Prediction
Rep3: 2000 T-ITS | A probabilistic approach to aircraft conflict detection

Topic 66 | grade A | fails: none | n=891

NPMI 0.022 | Jaccard 1.000 | modularity 0.059
Top-10: connected, vehicles, mixed, automated, traffic, autonomous, lane, cooperative, merging, dedicated
Journal: T-ITS 0.26 | peak 2022 | slope +0.079pp/yr | nonfixed 0.21
Rep1: 2016 TR-C | Influence of connected and autonomous vehicles on traffic flow stability and throughput
Rep2: 2017 TR-B | A mixed traffic capacity analysis and lane management model for connected automated vehicles: A Markov chain m
Rep3: 2017 TR-C | Eco approaching at an isolated signalized intersection under partially connected and automated vehicles enviro

Topic 67 | grade A | fails: none | n=834

NPMI -0.043 | Jaccard 0.818 | modularity 0.104
Top-10: driving, driver, cognitive, drivers, distraction, fatigue, workload, eye, task, visual
Journal: TR-F 0.35 | peak 2011 | slope +0.014pp/yr | nonfixed 0.07
Rep1: 2022 TR-F | Cognitive load, working memory capacity and driving performance: A preliminary fNIRS and eye tracking study
Rep2: 2016 AAP | Is take-over time all that matters? The impact of visual-cognitive load on driver take-over quality after cond
Rep3: 2005 AAP | Driver distraction: The effects of concurrent in-vehicle tasks, road environment complexity and age on driving

Topic 68 | grade A | fails: none | n=336

NPMI 0.021 | Jaccard 0.818 | modularity 0.084
Top-10: control, formation, triggered, tracking, unmanned, surface, adaptive, event, attacks, prescribed
Journal: T-ITS 0.43 | peak 2024 | slope +0.034pp/yr | nonfixed 0.46
Rep1: 2014 TR-C | Modeling cooperative and autonomous adaptive cruise control dynamic responses using experimental data
Rep2: 2008 TR-C | Adaptive cruise control design for active congestion avoidance
Rep3: 2021 T-ITS | Event-Triggered Adaptive Neural Fault-Tolerant Control of Underactuated MSVs With Input Saturation

Topic 69 | grade A | fails: none | n=783

NPMI 0.016 | Jaccard 0.818 | modularity 0.076
Top-10: prediction, graph, traffic, temporal, forecasting, flow, convolutional, network, learning, deep
Journal: T-ITS 0.60 | peak 2025 | slope +0.085pp/yr | nonfixed 0.09
Rep1: 2019 T-ITS | T-GCN: A Temporal Graph Convolutional Network for Traffic Prediction
Rep2: 2014 T-ITS | Traffic Flow Prediction With Big Data: A Deep Learning Approach
Rep3: 2017 TR-C | Deep learning for short-term traffic flow prediction

Topic 70 | grade A | fails: none | n=633

NPMI -0.041 | Jaccard 0.818 | modularity 0.089
Top-10: freight, intermodal, multimodal, transportation, logistics, network, transport, modal, demand, model
Journal: TRR 0.28 | peak 2004 | slope -0.017pp/yr | nonfixed 0.16
Rep1: 2006 TR-D | Modelling the full costs of an intermodal and road freight transport network
Rep2: 2020 TranspPol | The integration of passenger and freight transport for first-last mile operations
Rep3: 2021 TR-E | Urban rail service design for collaborative passenger and freight transport

Topic 71 | grade A | fails: none | n=581

NPMI -0.103 | Jaccard 0.818 | modularity 0.038
Top-10: covid, 19, pandemic, travel, post, mobility, public, spread, changes, ridership
Journal: TRR 0.20 | peak 2022 | slope +0.062pp/yr | nonfixed 0.33
Rep1: 2021 TranspPol | Public transit use in the United States in the era of COVID-19: Transit riders' travel behavior in the COVID-1
Rep2: 2021 JTG | The impact of the COVID-19 pandemic on people's mobility: A longitudinal study of the U.S. from March to Septe
Rep3: 2020 TranspPol | Understanding public transport satisfaction in post COVID-19 pandemic

Topic 72 | grade A | fails: none | n=788

NPMI -0.033 | Jaccard 0.818 | modularity 0.183
Top-10: transportation, projects, project, highway, construction, asset, management, agencies, state, practices
Journal: TRR 0.90 | peak 2004 | slope -0.134pp/yr | nonfixed 0.03
Rep1: 2022 TRR | Mapping of State Transportation Agencies' Practices and Perceptions in Project Bundling
Rep2: 2021 TRR | Equity-Oriented Criteria for Project Prioritization in Regional Transportation Planning
Rep3: 2004 TRR | Methodology for Multicriteria Decision Making in Highway Asset Management

Topic 73 | grade A | fails: none | n=658

NPMI -0.095 | Jaccard 0.818 | modularity 0.163
Top-10: ownership, car, built, residential, commuting, environment, household, housing, households, automobile
Journal: JTG 0.20 | peak 2017 | slope +0.017pp/yr | nonfixed 0.14
Rep1: 2017 TR-A | Exploring the influence of built environment on travel mode choice considering the mediating effects of car ow
Rep2: 2017 JTG | Residential self-selection and travel behaviour: What are the effects of attitudes, reasons for location choic
Rep3: 2006 TR-B | A comprehensive analysis of built environment characteristics on household residential choice and auto ownersh

Topic 74 | grade A | fails: none | n=620

NPMI -0.069 | Jaccard 0.818 | modularity 0.180
Top-10: airport, airline, airports, air, airlines, competition, market, aviation, domestic, international
Journal: TR-A 0.19 | peak 2010 | slope -0.034pp/yr | nonfixed 0.09
Rep1: 2021 RTBM | Airport service quality and passenger satisfaction: The impact of service failure on the likelihood of promoti
Rep2: 2015 TR-A | Low cost carrier competition and route entry in an emerging but regulated aviation market – The case of China
Rep3: 2017 TR-A | Vulnerability of the European air transport network to major airport closures from the perspective of passenge

Topic 75 | grade B | fails: F2 | n=989

NPMI -0.202 | Jaccard 0.667 | modularity 0.077
Top-10: travel, satisfaction, behaviour, mode, behavior, social, commute, surveys, activity, active
Journal: TR-A 0.20 | peak 2014 | slope -0.000pp/yr | nonfixed 0.18
Rep1: 2022 TR-F | The Long-Term effects of COVID-19 on travel behavior in the United States: A panel study on work from home, mo
Rep2: 2016 TR-D | Satisfaction with the commute: The role of travel mode choice, built environment and attitudes
Rep3: 2002 TRR | Mobility Styles and Travel Behavior: Application of a Lifestyle Approach to Leisure Travel

Topic 76 | grade B | fails: F2 | n=452

NPMI -0.429 | Jaccard 0.667 | modularity 0.228
Top-10: market, markets, sourcing, pricing, credit, competition, contracts, shipping, tradable, financing
Journal: TR-E 0.44 | peak 2017 | slope +0.015pp/yr | nonfixed 0.03
Rep1: 2020 TR-B | Pricing and equilibrium in on-demand ride-pooling markets
Rep2: 2016 TR-C | Economic analysis of ride-sourcing markets
Rep3: 2021 TR-B | Competition and third-party platform-integration in ride-sourcing markets

Topic 77 | grade A | fails: none | n=559

NPMI -0.029 | Jaccard 0.818 | modularity 0.067
Top-10: planning, trajectory, autonomous, motion, avoidance, vehicles, control, path, predictive, collision
Journal: T-ITS 0.44 | peak 2023 | slope +0.056pp/yr | nonfixed 0.34
Rep1: 2016 T-ITS | A Potential Field-Based Model Predictive Path-Planning Controller for Autonomous Road Vehicles
Rep2: 2022 T-ITS | Autonomous Driving on Curvy Roads Without Reliance on Frenet Frame: A Cartesian-Based Trajectory Planning Meth
Rep3: 2019 T-IV | Autonomous Driving Motion Planning With Constrained Iterative LQR

Topic 78 | grade A | fails: none | n=825

NPMI -0.100 | Jaccard 1.000 | modularity 0.131
Top-10: driving, driver, drivers, naturalistic, simulator, training, behavior, hazard, novice, behaviour
Journal: AAP 0.38 | peak 2016 | slope +0.020pp/yr | nonfixed 0.07
Rep1: 2020 AAP | Analyzing driver behavior under naturalistic driving conditions: A review
Rep2: 2020 AAP | The impact of the connected environment on driving behavior and safety: A driving simulator study
Rep3: 2017 TR-F | Why is steering not the same as braking? The impact of non-driving related tasks on lateral and longitudinal d

Topic 79 | grade A | fails: none | n=594

NPMI -0.094 | Jaccard 0.667 | modularity 0.129
Top-10: problem, routing, drone, delivery, location, drones, optimization, shortest, robust, multi
Journal: TR-E 0.26 | peak 2016 | slope +0.027pp/yr | nonfixed 0.05
Rep1: 2019 TR-C | An adaptive large neighborhood search metaheuristic for the vehicle routing problem with drones
Rep2: 2014 TR-E | A heuristic approach for the green vehicle routing problem with multiple technologies and partial recharges
Rep3: 2015 TR-C | The flying sidekick traveling salesman problem: Optimization of drone-assisted parcel delivery

Topic 80 | grade A | fails: none | n=674

NPMI 0.071 | Jaccard 0.818 | modularity 0.091
Top-10: problem, routing, problems, windows, branch, delivery, cut, algorithm, inventory, vehicle
Journal: TransSci 0.54 | peak 2003 | slope -0.024pp/yr | nonfixed 0.01
Rep1: 2004 TransSci | A Two-Stage Hybrid Local Search for the Vehicle Routing Problem with Time Windows
Rep2: 2005 TransSci | Vehicle Routing Problem with Time Windows, Part I: Route Construction and Local Search Algorithms
Rep3: 2019 TransSci | The Robust Vehicle Routing Problem with Time Windows: Compact Formulation and Branch-Price-and-Cut Method

Topic 81 | grade A | fails: none | n=438

NPMI -0.148 | Jaccard 0.667 | modularity 0.203
Top-10: logistics, supply, industry, chain, innovation, firms, companies, green, providers, firm
Journal: TR-E 0.34 | peak 2017 | slope +0.006pp/yr | nonfixed 0.15
Rep1: 2017 TR-E | Information technology for competitive advantage within logistics and supply chains: A review
Rep2: 2013 TR-E | The effect of green supply chain management on green performance and firm competitiveness in the context of co
Rep3: 2014 TRR | Logistics Sprawl

Topic 82 | grade B | fails: F2 | n=501

NPMI -0.326 | Jaccard 0.818 | modularity 0.195
Top-10: evacuation, emergency, hurricane, disaster, disasters, relief, response, medical, crowd, evacuations
Journal: TRR 0.33 | peak 2009 | slope +0.011pp/yr | nonfixed 0.08
Rep1: 2009 TR-E | Dynamic relief-demand management for emergency logistics operations under large-scale disasters
Rep2: 2009 TR-B | Pre-positioning of emergency supplies for disaster response
Rep3: 2015 TR-E | Modeling multiple humanitarian objectives in emergency response to large-scale disasters

Topic 83 | grade B | fails: F2 | n=833

NPMI -0.263 | Jaccard 0.818 | modularity 0.109
Top-10: safety, road, surrogate, crash, risk, intersections, functions, collision, crossings, accident
Journal: AAP 0.44 | peak 2013 | slope -0.029pp/yr | nonfixed 0.07
Rep1: 2001 AAP | Extended time-to-collision measures for road traffic safety assessment
Rep2: 2022 TR-C | Anticipated Collision Time (ACT): A two-dimensional surrogate safety indicator for trajectory-based proactive
Rep3: 2014 AAP | Simulation of safety: A review of the state of the art in road safety simulation modelling

Topic 84 | grade A | fails: none | n=309

NPMI -0.056 | Jaccard 0.667 | modularity 0.139
Top-10: bridge, steel, bridges, concrete, girder, reinforced, load, deck, prestressed, girders
Journal: TRR 0.92 | peak 2004 | slope -0.083pp/yr | nonfixed 0.04
Rep1: 2015 TRR | Load Rating a Prestressed Concrete Double T-Beam Bridge without Plans by Field Testing
Rep2: 2018 TRR | New Connection Detail to Connect Precast Column to Cap Beam using Ultra-High-Performance Concrete in Accelerat
Rep3: 2014 TRR | Design of Ultrahigh-Performance Concrete Waffle Deck for Accelerated Bridge Construction

Topic 85 | grade A | fails: none | n=333

NPMI -0.195 | Jaccard 0.667 | modularity 0.131
Top-10: diagnosis, fault, railway, traction, track, anomaly, monitoring, detection, train, inspection
Journal: T-ITS 0.28 | peak 2024 | slope +0.025pp/yr | nonfixed 0.34
Rep1: 2020 T-ITS | Data-Driven Fault Diagnosis for Traction Systems in High-Speed Trains: A Survey, Challenges, and Perspectives
Rep2: 2021 T-TE | Artificial Intelligence-Based Technique for Fault Detection and Diagnosis of EV Motors: A Review
Rep3: 2019 T-ITS | A Review of Fault Detection and Diagnosis for the Traction System in High-Speed Trains

Topic 86 | grade A | fails: none | n=388

NPMI -0.170 | Jaccard 0.818 | modularity 0.150
Top-10: injury, injuries, head, occupant, frontal, collisions, crashes, crash, accidents, kinematics
Journal: AAP 0.83 | peak 2000 | slope -0.041pp/yr | nonfixed 0.03
Rep1: 2020 AAP | Impact speed and the risk of serious injury in vehicle crashes
Rep2: 2012 AAP | Incidence trends of traumatic spinal cord injury and traumatic brain injury in Spain, 2000–2009
Rep3: 2013 AAP | Angular Impact Mitigation system for bicycle helmets to reduce head acceleration and risk of traumatic brain i

Topic 87 | grade A | fails: none | n=809

NPMI -0.019 | Jaccard 0.818 | modularity 0.067
Top-10: choice, activity, travel, mode, route, modeling, choices, discrete, trips, latent
Journal: TRR 0.46 | peak 2005 | slope -0.058pp/yr | nonfixed 0.12
Rep1: 2018 TRR | Comparison of Four Types of Artificial Neural Network and a Multinomial Logit Model for Travel Mode Choice Mod
Rep2: 2013 TRR | Hybrid Choice Model to Investigate Effects of Teenagers' Attitudes toward Walking and Cycling on Mode Choice B
Rep3: 2013 TR-A | Incorporating the influence of latent modal preferences on travel mode choice behavior

Topic 88 | grade A | fails: none | n=653

NPMI -0.012 | Jaccard 0.818 | modularity 0.099
Top-10: permanent, magnet, motor, flux, machines, machine, synchronous, magnetic, torque, rotor
Journal: T-TE 0.99 | peak 2025 | slope +0.073pp/yr | nonfixed 1.00
Rep1: 2022 T-TE | Sequential Subspace Optimization Design of a Dual Three-Phase Permanent Magnet Synchronous Hub Motor Based on
Rep2: 2022 T-TE | Performance Analysis of Magnetically Geared Permanent Magnet Brushless Motor for Hybrid Electric Vehicles
Rep3: 2015 T-TE | Comparative Study of Interior Permanent Magnet, Induction, and Switched Reluctance Motor Drives for EV and HEV

Topic 89 | grade A | fails: none | n=371

NPMI -0.193 | Jaccard 0.818 | modularity 0.128
Top-10: pavement, friction, roughness, surface, texture, measurement, skid, measurements, laser, tire
Journal: TRR 0.72 | peak 2002 | slope -0.018pp/yr | nonfixed 0.13
Rep1: 2014 TRR | Measurement of Pavement Roughness Using Android-Based Smartphone Application
Rep2: 2017 TRR | Novel Macro- and Microtexture Indicators for Pavement Friction by Using High-Resolution Three-Dimensional Surf
Rep3: 2003 TRR | Pavement Surface Macrotexture Measurement and Applications

Topic 90 | grade B | fails: F2 | n=418

NPMI -0.223 | Jaccard 0.667 | modularity 0.263
Top-10: reliability, uncertainty, risk, time, maintenance, probabilistic, travel, railroad, stochastic, bayesian
Journal: TRR 0.30 | peak 2013 | slope -0.012pp/yr | nonfixed 0.06
Rep1: 2006 TRR | Reliable Network Design Problem: Case with Uncertain Demand and Total Travel Time Reliability
Rep2: 2022 TR-C | Travel time reliability in transportation networks: A review of methodological developments
Rep3: 2002 TRR | Travel Time Reliability with Risk-Sensitive Travelers

Topic 91 | grade A | fails: none | n=168

NPMI -0.068 | Jaccard 0.667 | modularity 0.086
Top-10: reinforcement, learning, deep, auv, robot, uav, path, multi, planning, agent
Journal: T-ITS 0.48 | peak 2024 | slope +0.019pp/yr | nonfixed 0.39
Rep1: 2020 T-IV | Comparison of Deep Reinforcement Learning and Model Predictive Control for Adaptive Cruise Control
Rep2: 2020 T-ITS | Path Following Optimization for an Underactuated USV Using Smoothly-Convergent Deep Reinforcement Learning
Rep3: 2021 T-ITS | Reinforcement Learning and Particle Swarm Optimization Supporting Real-Time Rescue Assignments for Multiple Au

Topic 92 | grade A | fails: none | n=801

NPMI -0.148 | Jaccard 0.818 | modularity 0.114
Top-10: transit, bus, rapid, ridership, service, rail, dwell, york, public, fare
Journal: TRR 0.54 | peak 2013 | slope -0.034pp/yr | nonfixed 0.07
Rep1: 2014 TranspPol | BRT TOD: Leveraging transit oriented development with bus rapid transit investments
Rep2: 2008 TRR | Bus Rapid Transit Systems in Latin America and Asia
Rep3: 2002 JPT | Bus Rapid Transit: An Overview

Topic 93 | grade A | fails: none | n=269

NPMI -0.192 | Jaccard 0.667 | modularity 0.167
Top-10: pavement, quality, assurance, specifications, construction, performance, condition, maintenance, distress, pavements
Journal: TRR 0.96 | peak 2002 | slope -0.072pp/yr | nonfixed 0.02
Rep1: 2008 TRR | Review of Louisiana's Pavement Management System
Rep2: 2006 TRR | Enhancement of Arizona Pavement Management System for Construction and Maintenance Activities
Rep3: 2003 TRR | Pavement Management Perspective on Integrating Preventive Maintenance into a Pavement Management System

Topic 94 | grade A | fails: none | n=728

NPMI -0.040 | Jaccard 0.667 | modularity 0.126
Top-10: transport, mobility, sustainable, policy, governance, public, planning, cities, justice, urban
Journal: TranspPol 0.21 | peak 2013 | slope +0.012pp/yr | nonfixed 0.27
Rep1: 2012 TranspPol | Transport and social exclusion: Where are we now?
Rep2: 2017 ETRR | Crowd logistics: an opportunity for more sustainable urban freight transport?
Rep3: 2013 JTG | Why culture matters for transport policy: the case of cycling in the UK

Topic 95 | grade B | fails: F2 | n=448

NPMI -0.211 | Jaccard 0.667 | modularity 0.090
Top-10: supply, chain, chains, product, inventory, green, closed, supplier, production, carbon
Journal: TR-E 0.79 | peak 2015 | slope +0.023pp/yr | nonfixed 0.02
Rep1: 2022 TR-E | Cloud supply chain: Integrating Industry 4.0 and digital platforms in the “Supply Chain-as-a-Service”
Rep2: 2016 TR-E | Marrying supply chain sustainability and resilience: A match made in heaven
Rep3: 2016 TR-E | Sustainable design of a closed-loop location-routing-inventory supply chain network under mixed uncertainty

Topic 96 | grade B | fails: F2 | n=333

NPMI -0.256 | Jaccard 0.538 | modularity 0.262
Top-10: soil, soils, pile, piles, earth, stabilized, reinforced, penetration, test, pipe
Journal: TRR 0.84 | peak 2001 | slope -0.066pp/yr | nonfixed 0.05
Rep1: 2007 TRR | Laboratory Investigation of Behavior of Foundations on Geosynthetic-Reinforced Clayey Soil
Rep2: 2020 TRR | Exploring Artificial Neural Network to Evaluate the Undrained Shear Strength of Soil from Cone Penetration Tes
Rep3: 2013 TRR | Reinforcement Pullout Capacity in Mechanically Stabilized Earth Walls with Marginal-Quality Soils

Topic 97 | grade A | fails: none | n=562

NPMI 0.033 | Jaccard 0.818 | modularity 0.101
Top-10: satisfaction, service, quality, public, passengers, customer, loyalty, perceived, bus, services
Journal: TRR 0.26 | peak 2005 | slope -0.018pp/yr | nonfixed 0.18
Rep1: 2013 TranspPol | Perceived service quality in bus transit service: A structural equation approach
Rep2: 2008 TR-A | Investigating structural relationships between service quality, perceived value, satisfaction, and behavioral
Rep3: 2010 TranspPol | Behavioral intentions of public transit passengers—The roles of service quality, perceived value, satisfaction

Topic 98 | grade A | fails: none | n=644

NPMI -0.025 | Jaccard 0.818 | modularity 0.204
Top-10: container, problem, scheduling, allocation, terminals, berth, crane, yard, delivery, terminal
Journal: TR-E 0.42 | peak 2015 | slope +0.019pp/yr | nonfixed 0.03
Rep1: 2015 TR-E | Integrated Berth Allocation and Quay Crane Assignment Problem: Set partitioning models and computational resul
Rep2: 2001 TR-B | The dynamic berth allocation problem for a container port
Rep3: 2006 TR-E | Quay crane scheduling with non-interference constraints in port container terminals

Topic 99 | grade A | fails: none | n=498

NPMI 0.052 | Jaccard 0.818 | modularity 0.086
Top-10: vehicular, edge, computing, offloading, internet, resource, networks, intelligent, computation, federated
Journal: T-ITS 0.71 | peak 2021 | slope +0.051pp/yr | nonfixed 0.18
Rep1: 2022 T-ITS | Mobility-Aware Multi-Hop Task Offloading for Autonomous Driving in Vehicular Edge Computing and Networks
Rep2: 2021 T-ITS | A Taxonomy and Survey of Edge Cloud Computing for Intelligent Transportation Systems and Connected Vehicles
Rep3: 2020 T-ITS | Intelligent Edge Computing in Internet of Vehicles: A Joint Computation Offloading and Caching Solution

Topic 100 | grade A | fails: none | n=495

NPMI -0.149 | Jaccard 0.818 | modularity 0.145
Top-10: choice, discrete, logit, models, stated, utility, model, latent, multinomial, continuous
Journal: TR-B 0.39 | peak 2011 | slope -0.022pp/yr | nonfixed 0.05
Rep1: 2003 TR-B | A latent class model for discrete choice analysis: contrasts with mixed logit
Rep2: 2010 TRR | Sequential and Simultaneous Estimation of Hybrid Discrete Choice Models
Rep3: 2020 TR-B | Enhancing discrete choice models with representation learning

Topic 101 | grade A | fails: none | n=578

NPMI -0.078 | Jaccard 0.667 | modularity 0.113
Top-10: land, spatial, use, urban, development, transportation, firm, location, form, metropolitan
Journal: TRR 0.33 | peak 2007 | slope -0.042pp/yr | nonfixed 0.09
Rep1: 2014 JTG | How land-use and urban form impact bicycle flows: evidence from the bicycle-sharing system (BIXI) in Montreal
Rep2: 2005 TranspRev | Current operational urban land-use–transport modelling frameworks: A review
Rep3: 2014 TRR | Synthetic Population Generation at Disaggregated Spatial Scales for Land Use and Transportation Microsimulatio

Topic 102 | grade B | fails: F2 | n=620

NPMI -0.259 | Jaccard 0.742 | modularity 0.163
Top-10: pavement, maintenance, pavements, mechanistic, life, rehabilitation, cycle, condition, guide, deterioration
Journal: TRR 0.77 | peak 2006 | slope -0.051pp/yr | nonfixed 0.07
Rep1: 2009 TRR | Calibration of Flexible Pavement in Mechanistic–Empirical Pavement Design Guide for Washington State
Rep2: 2012 TRR | Stochastic Life-Cycle Cost Analysis for Pavement Preservation Treatments
Rep3: 2000 TRR | Incorporation of Reliability into Mechanistic-Empirical Pavement Design

Topic 103 | grade A | fails: none | n=403

NPMI -0.042 | Jaccard 0.818 | modularity 0.073
Top-10: asphalt, recycled, pavement, mixtures, reclaimed, binder, cold, recycling, mix, concrete
Journal: TRR 0.88 | peak 2011 | slope +0.002pp/yr | nonfixed 0.08
Rep1: 2005 TRR | Laboratory Investigation of Mixing Hot-Mix Asphalt with Reclaimed Asphalt Pavement
Rep2: 2013 TRR | Evaluation of Rejuvenator's Effectiveness with Conventional Mix Testing for 100% Reclaimed Asphalt Pavement Mi
Rep3: 2011 TRR | Effects of Recycled Asphalt Pavement Amounts on Low-Temperature Cracking Performance of Asphalt Mixtures Using

Topic 104 | grade A | fails: none | n=720

NPMI 0.001 | Jaccard 0.818 | modularity 0.086
Top-10: equilibrium, assignment, user, dynamic, stochastic, network, traffic, networks, problem, link
Journal: TR-B 0.39 | peak 2000 | slope -0.051pp/yr | nonfixed 0.01
Rep1: 2012 TRR | Path-Based Algorithms to Solve C-Logit Stochastic User Equilibrium Assignment Problem
Rep2: 2006 TR-B | A path-based user-equilibrium traffic assignment algorithm that obviates path storage and enumeration
Rep3: 2015 TR-B | Formulation, existence, and computation of boundedly rational dynamic user equilibrium with fixed or endogenou

Topic 105 | grade A | fails: none | n=517

NPMI -0.177 | Jaccard 0.667 | modularity 0.094
Top-10: lidar, detection, tracking, roadside, map, point, camera, vision, 3d, clouds
Journal: T-ITS 0.54 | peak 2008 | slope +0.016pp/yr | nonfixed 0.21
Rep1: 2013 T-ITS | Looking at Vehicles on the Road: A Survey of Vision-Based Vehicle Detection, Tracking, and Behavior Analysis
Rep2: 2019 TR-C | Detection and tracking of pedestrians and vehicles using roadside LiDAR sensors
Rep3: 2018 TRR | Automatic Background Filtering Method for Roadside LiDAR Data

Topic 106 | grade A | fails: none | n=399

NPMI -0.051 | Jaccard 0.818 | modularity 0.058
Top-10: asphalt, mix, hot, mixtures, warm, superpave, laboratory, moisture, mixture, performance
Journal: TRR 0.99 | peak 2004 | slope -0.078pp/yr | nonfixed 0.01
Rep1: 2012 TRR | Laboratory Performance Evaluation of Warm-Mix Asphalt Containing High Percentages of Reclaimed Asphalt Pavemen
Rep2: 2007 TRR | Field Performance of Warm-Mix Asphalt at National Center for Asphalt Technology Test Track
Rep3: 2016 TRR | Long-Term Field Rutting and Moisture Susceptibility Performance of Warm-Mix Asphalt Pavement

Topic 107 | grade A | fails: none | n=735

NPMI -0.089 | Jaccard 0.818 | modularity 0.187
Top-10: signal, intersections, signalized, priority, intersection, timing, control, traffic, arterial, delay
Journal: TRR 0.64 | peak 2000 | slope -0.074pp/yr | nonfixed 0.06
Rep1: 2013 TRR | Traffic Signal Control with Connected Vehicles
Rep2: 2011 TRR | Traffic Signal Optimization with Application of Transit Signal Priority to an Isolated Intersection
Rep3: 2011 TRR | Heuristic Algorithm for Priority Traffic Signal Control

Topic 108 | grade B | fails: F2 | n=334

NPMI -0.486 | Jaccard 0.818 | modularity 0.200
Top-10: visual, visibility, lighting, light, glance, red, eye, night, cameras, conspicuity
Journal: TRR 0.32 | peak 2007 | slope -0.015pp/yr | nonfixed 0.06
Rep1: 2012 AAP | Driving in fog: The effects of driving experience and visibility on speed compensation and hazard avoidance
Rep2: 2005 TRR | Eye Glance Behavior of Van and Passenger Car Drivers During Lane Change Decision Phase
Rep3: 2005 TRR | Eye Glance Behavior of van and Passenger Car Drivers during Lane Change Decision Phase

Topic 109 | grade B | fails: F2 | n=419

NPMI -0.198 | Jaccard 0.818 | modularity 0.199
Top-10: asphalt, modified, cement, binders, properties, concrete, binder, aging, rubber, ash
Journal: TRR 0.82 | peak 2011 | slope -0.015pp/yr | nonfixed 0.14
Rep1: 2015 TRR | Effect of Aging on Adhesion Properties of Asphalt Mixtures with the Use of Bitumen Bond Strength and Surface E
Rep2: 2014 TRR | Laboratory Investigation of Biochar-Modified Asphalt Mixture
Rep3: 2018 TRR | Rheological Behavior and Its Chemical Interpretation of Crumb Rubber Modified Asphalt Containing Warm-Mix Addi

Topic 110 | grade A | fails: none | n=613

NPMI -0.041 | Jaccard 0.667 | modularity 0.087
Top-10: traffic, estimation, data, probe, flow, freeway, queue, detector, signalized, fundamental
Journal: TRR 0.55 | peak 2010 | slope -0.022pp/yr | nonfixed 0.07
Rep1: 2003 TRR | Application of Probe-Vehicle Data for Real-Time Traffic-State Estimation and Short-Term Travel-Time Prediction
Rep2: 2009 T-ITS | PPCA-Based Missing Data Imputation for Traffic Flow Volume: A Systematical Approach
Rep3: 2017 TRR | Real-Time Queue Length Estimation for Signalized Intersections Using Vehicle Trajectory Data

Topic 111 | grade A | fails: none | n=598

NPMI -0.181 | Jaccard 1.000 | modularity 0.157
Top-10: airport, air, flight, aircraft, airline, airspace, aviation, delay, operations, airports
Journal: TRR 0.24 | peak 2002 | slope -0.014pp/yr | nonfixed 0.11
Rep1: 2011 TR-C | Modelling delay propagation within an airport network
Rep2: 2020 TransSci | Dynamic Disruption Management in Airline Networks Under Airport Operating Uncertainty
Rep3: 2010 TRR | Flight Delay Propagation Impact on Strategic Air Traffic Flow Management

Topic 112 | grade A | fails: none | n=297

NPMI -0.121 | Jaccard 0.538 | modularity 0.039
Top-10: reinforcement, learning, deep, agent, multi, scheduling, delivery, dispatching, critic, metering
Journal: T-ITS 0.39 | peak 2025 | slope +0.032pp/yr | nonfixed 0.11
Rep1: 2022 TR-C | Reinforcement learning for ridesharing: An extended survey
Rep2: 2021 T-ITS | Deep Reinforcement Learning for the Electric Vehicle Routing Problem With Time Windows
Rep3: 2019 T-ITS | Online Vehicle Routing With Neural Combinatorial Optimization and Deep Reinforcement Learning

Topic 113 | grade A | fails: none | n=418

NPMI -0.059 | Jaccard 0.818 | modularity 0.094
Top-10: crack, detection, pavement, segmentation, images, deep, defect, distress, neural, learning
Journal: CACIE 0.48 | peak 2025 | slope +0.039pp/yr | nonfixed 0.04
Rep1: 2019 T-ITS | Feature Pyramid and Hierarchical Boosting Network for Pavement Crack Detection
Rep2: 2022 T-ITS | DMA-Net: DeepLab With Multi-Scale Attention for Pavement Crack Segmentation
Rep3: 2020 CACIE | Automated pavement crack detection and segmentation based on two-step convolutional neural network

Topic 114 | grade B | fails: F2 | n=566

NPMI -0.259 | Jaccard 0.818 | modularity 0.154
Top-10: emissions, air, pollution, emission, exposure, concentrations, pollutant, quality, noise, particulate
Journal: TR-D 0.55 | peak 2017 | slope -0.004pp/yr | nonfixed 0.08
Rep1: 2013 TRR | Vehicle Emission and Near-Road Air Quality Modeling for Shanghai, China
Rep2: 2014 TRR | Dispersion Modeling of Traffic-Related Air Pollutant Exposures and Health Effects among Children with Asthma i
Rep3: 2015 TR-B | Urban transportation emissions mitigation: Coupling high-resolution vehicular emissions and traffic models for

Topic 115 | grade B | fails: F2 | n=276

NPMI -0.290 | Jaccard 0.667 | modularity 0.138
Top-10: battery, electric, energy, storage, power, electronics, hybrid, cell, vehicles, automotive
Journal: T-TE 0.37 | peak 2023 | slope +0.028pp/yr | nonfixed 0.69
Rep1: 2017 TR-B | Battery degradation and behaviour for electric vehicles: Review and numerical analyses of several models
Rep2: 2022 GEIT | China's battery electric vehicles lead the world: achievements in technology system architecture and technolog
Rep3: 2014 TR-D | Electric vehicles' energy consumption measurement and estimation

Topic 116 | grade A | fails: none | n=420

NPMI -0.141 | Jaccard 0.667 | modularity 0.053
Top-10: transit, design, assignment, network, feeder, bus, demand, optimal, frequency, schedule
Journal: TRR 0.24 | peak 2002 | slope -0.021pp/yr | nonfixed 0.04
Rep1: 2021 TR-B | Integrated line configuration and frequency determination with passenger path assignment in urban rail transit
Rep2: 2006 TRR | Minimizing Transfer Times in Public Transit Network with Genetic Algorithm
Rep3: 2010 TR-C | Transit network design: A procedure and an application to a large urban area

Topic 117 | grade B | fails: F2 | n=318

NPMI -0.507 | Jaccard 0.667 | modularity 0.278
Top-10: enforcement, law, licensing, laws, violations, belt, legal, graduated, speeding, offenders
Journal: AAP 0.49 | peak 2005 | slope -0.028pp/yr | nonfixed 0.05
Rep1: 2013 TRR | Effects of High-Visibility Enforcement on Driver Compliance with Pedestrian Yield Right-of-Way Laws
Rep2: 2005 AAP | The effectiveness of enforcement and publicity campaigns on serious crashes involving young male drivers: Are
Rep3: 2000 AAP | Effect of Florida's graduated licensing program on the crash rate of teenage drivers

Topic 118 | grade A | fails: none | n=452

NPMI -0.070 | Jaccard 0.818 | modularity 0.123
Top-10: electric, adoption, consumer, consumers, purchase, car, preferences, vehicles, fuel, vehicle
Journal: TR-D 0.23 | peak 2018 | slope +0.021pp/yr | nonfixed 0.17
Rep1: 2012 TR-A | The role of instrumental, hedonic and symbolic attributes in the intention to adopt electric vehicles
Rep2: 2022 TR-D | Consumers' preferences for electric vehicles: The role of status and reputation
Rep3: 2018 TR-A | Policy implications for promoting the adoption of electric vehicles: Do consumer's knowledge, perceived risk a

Topic 119 | grade A | fails: none | n=422

NPMI -0.146 | Jaccard 0.818 | modularity 0.139
Top-10: flow, macroscopic, traffic, model, continuum, fundamental, microscopic, kinematic, pedestrian, dynamics
Journal: TR-B 0.30 | peak 2002 | slope -0.024pp/yr | nonfixed 0.06
Rep1: 2002 TRR | Kinematic Wave Traffic Flow Model for Mixed Traffic
Rep2: 2003 TRR | Two-Phase Bounded-Acceleration Traffic Flow Model: Analytical Solutions and Applications
Rep3: 2002 TR-B | A new continuum model for traffic flow and numerical tests

Topic 120 | grade A | fails: none | n=349

NPMI -0.042 | Jaccard 1.000 | modularity 0.078
Top-10: ion, lithium, battery, batteries, state, estimation, health, charge, degradation, diagnosis
Journal: T-TE 0.71 | peak 2025 | slope +0.039pp/yr | nonfixed 0.98
Rep1: 2022 T-TE | Electrochemical Impedance Spectroscopy Based State-of-Health Estimation for Lithium-Ion Battery Considering Te
Rep2: 2021 T-TE | A Flexible State-of-Health Prediction Scheme for Lithium-Ion Battery Packs With Long Short-Term Memory Network
Rep3: 2019 T-TE | Modified Gaussian Process Regression Models for Cyclic Capacity Prediction of Lithium-Ion Batteries

Topic 121 | grade A | fails: none | n=581

NPMI -0.151 | Jaccard 0.818 | modularity 0.097
Top-10: port, ports, container, hinterland, governance, inland, competition, sea, maritime, shipping
Journal: JTG 0.22 | peak 2017 | slope +0.017pp/yr | nonfixed 0.19
Rep1: 2019 IJST | A framework for building a smart port and smart port index
Rep2: 2008 JTG | The dry port concept: connecting container seaports with the hinterland
Rep3: 2017 RTBM | Revisiting port governance and port reform: A multi-country examination

Topic 122 | grade A | fails: none | n=806

NPMI -0.052 | Jaccard 0.818 | modularity 0.103
Top-10: driving, driver, automated, drivers, automation, simulator, visual, awareness, situation, interface
Journal: TR-F 0.39 | peak 2020 | slope +0.034pp/yr | nonfixed 0.14
Rep1: 2018 TR-F | Take-over requests in highly automated driving: A crowdsourcing survey on auditory, vibrotactile, and visual d
Rep2: 2020 ETRR | Multimodal warning design for take-over request in conditionally automated driving
Rep3: 2021 AAP | Takeover requests for automated driving: The effects of signal direction, lead time, and modality on takeover

Topic 123 | grade A | fails: none | n=835

NPMI -0.093 | Jaccard 0.818 | modularity 0.133
Top-10: mobility, shared, maas, service, public, scooter, micromobility, transport, scooters, services
Journal: TRR 0.16 | peak 2025 | slope +0.067pp/yr | nonfixed 0.36
Rep1: 2016 TRR | Trying Out Mobility as a Service: Experiences from a Field Trial and Implications for Understanding Demand
Rep2: 2015 TRR | Implementing Mobility as a Service
Rep3: 2020 TR-F | Mobility as a service and sustainable travel behaviour: A thematic analysis study

Topic 124 | grade A | fails: none | n=478

NPMI -0.124 | Jaccard 0.818 | modularity 0.098
Top-10: shipping, maritime, ship, emissions, emission, ships, sea, container, oil, carbon
Journal: TR-D 0.40 | peak 2020 | slope +0.029pp/yr | nonfixed 0.13
Rep1: 2019 TR-D | Global impacts of recent IMO regulations on marine fuel oil refining processes and ship emissions
Rep2: 2011 TR-D | Is slow steaming a sustainable means of reducing CO2 emissions from container shipping?
Rep3: 2015 TRR | Environmental Balance of Shipping Emissions Reduction Strategies

Topic 125 | grade A | fails: none | n=544

NPMI -0.006 | Jaccard 0.818 | modularity 0.160
Top-10: power, dc, converter, wireless, voltage, transfer, current, ev, phase, bidirectional
Journal: T-TE 0.96 | peak 2023 | slope +0.060pp/yr | nonfixed 0.99
Rep1: 2016 T-TE | Comprehensive Analyses and Comparison of 1 kW Isolated DC–DC Converters for Bidirectional EV Charging Systems
Rep2: 2018 T-TE | A Bidirectional Soft-Switched DAB-Based Single-Stage Three-Phase AC–DC Converter for V2G Application
Rep3: 2020 T-TE | An Overview of High-Conversion High-Voltage DC–DC Converters for Electrified Aviation Power Distribution Syste

Topic 126 | grade A | fails: none | n=394

NPMI -0.194 | Jaccard 0.667 | modularity 0.289
Top-10: structural, bridge, structures, vibration, monitoring, identification, wavelet, damage, ultrasonic, seismic
Journal: CACIE 0.48 | peak 2000 | slope -0.030pp/yr | nonfixed 0.08
Rep1: 2006 CACIE | A New Approach for Health Monitoring of Structures: Terrestrial Laser Scanning
Rep2: 2005 CACIE | Dynamic Wavelet Neural Network for Nonlinear Identification of Highrise Buildings
Rep3: 2008 CACIE | Long-Term Monitoring and Identification of Bridge Structural Parameters

Topic 127 | grade B | fails: F2 | n=347

NPMI -0.269 | Jaccard 0.667 | modularity 0.222
Top-10: segmentation, semantic, scene, image, license, detection, scenes, plate, domain, rgb
Journal: T-ITS 0.71 | peak 2025 | slope +0.036pp/yr | nonfixed 0.15
Rep1: 2022 T-ITS | Deep Dual-Resolution Networks for Real-Time and Accurate Semantic Segmentation of Traffic Scenes
Rep2: 2017 T-ITS | ERFNet: Efficient Residual Factorized ConvNet for Real-Time Semantic Segmentation
Rep3: 2022 T-ITS | SFNet-N: An Improved SFNet Algorithm for Semantic Segmentation of Low-Light Autonomous Driving Road Scenes

Topic 128 | grade A | fails: none | n=505

NPMI -0.129 | Jaccard 0.818 | modularity 0.091
Top-10: scheduling, problem, aircraft, airline, crew, assignment, maintenance, timetabling, fleet, programming
Journal: TransSci 0.27 | peak 2001 | slope -0.040pp/yr | nonfixed 0.01
Rep1: 2001 TransSci | Benders Decomposition for Simultaneous Aircraft Routing and Crew Scheduling
Rep2: 2018 TRR | A Non-Dominated Sorting Genetic Algorithm Approach for Optimization of Multi-Objective Airport Gate Assignment
Rep3: 2002 TR-A | An optimization model for aircraft maintenance scheduling and re-assignment

Topic 129 | grade B | fails: F2 | n=415

NPMI -0.416 | Jaccard 0.667 | modularity 0.084
Top-10: parking, street, pricing, cruising, reservation, search, morning, shared, occupancy, curb
Journal: TR-A 0.16 | peak 2021 | slope +0.024pp/yr | nonfixed 0.13
Rep1: 2017 TRR | Parking Spaces in the Age of Shared Autonomous Vehicles: How Much Parking Will We Need and Where?
Rep2: 2006 TranspPol | Cruising for parking
Rep3: 2015 TRR | Parking Reservation for Managing Downtown Curbside Parking

Topic 130 | grade A | fails: none | n=288

NPMI -0.115 | Jaccard 0.667 | modularity 0.155
Top-10: uav, aerial, unmanned, assisted, uavs, enabled, path, trajectory, planning, resource
Journal: T-ITS 0.60 | peak 2024 | slope +0.030pp/yr | nonfixed 0.23
Rep1: 2021 T-ITS | A Clustering-Based Coverage Path Planning Method for Autonomous Heterogeneous UAVs
Rep2: 2015 VT Mag | Multi-UAV-Aided Networks: Aerial-Ground Cooperative Vehicular Networking Architecture
Rep3: 2022 T-ITS | Joint Communication and Trajectory Optimization for Multi-UAV Enabled Mobile Internet of Vehicles

Topic 131 | grade A | fails: none | n=494

NPMI -0.032 | Jaccard 0.818 | modularity 0.082
Top-10: reinforcement, learning, deep, autonomous, control, signal, agent, making, driving, decision
Journal: T-ITS 0.48 | peak 2025 | slope +0.054pp/yr | nonfixed 0.24
Rep1: 2021 T-ITS | Deep Reinforcement Learning for Autonomous Driving: A Survey
Rep2: 2021 TR-C | Decision making of autonomous vehicles in lane change scenarios: Deep reinforcement learning approaches with r
Rep3: 2019 T-ITS | Multi-Agent Deep Reinforcement Learning for Large-Scale Traffic Signal Control

Topic 132 | grade B | fails: F2 | n=464

NPMI -0.468 | Jaccard 0.818 | modularity 0.336
Top-10: child, injury, sleep, fatigue, restraint, belt, children, safety, seat, occupational
Journal: AAP 0.68 | peak 2010 | slope -0.031pp/yr | nonfixed 0.04
Rep1: 2011 AAP | Psychosocial safety climate as a lead indicator of workplace bullying and harassment, job resources, psycholog
Rep2: 2012 AAP | Job demands, job resources and safety outcomes: The roles of emotional exhaustion and safety compliance
Rep3: 2010 AAP | Modelling fatigue and the use of fatigue models in work settings

Topic 133 | grade A | fails: none | n=570

NPMI 0.028 | Jaccard 0.818 | modularity 0.120
Top-10: prediction, trajectory, intention, autonomous, driving, learning, graph, aware, transformer, attention
Journal: T-ITS 0.51 | peak 2025 | slope +0.063pp/yr | nonfixed 0.30
Rep1: 2022 T-IV | A Survey on Trajectory-Prediction Methods for Autonomous Driving
Rep2: 2020 T-ITS | Deep Learning-Based Vehicle Behavior Prediction for Autonomous Driving Applications: A Review
Rep3: 2022 T-ITS | Multi-Agent Trajectory Prediction With Heterogeneous Edge-Enhanced Graph Attention Network

Topic 134 | grade A | fails: none | n=479

NPMI -0.153 | Jaccard 0.667 | modularity 0.058
Top-10: resilience, vulnerability, network, transportation, infrastructure, assessment, disruptions, networks, disaster, flood
Journal: TRR 0.27 | peak 2025 | slope +0.037pp/yr | nonfixed 0.07
Rep1: 2018 TRR | Road Network Resilience: How to Identify Critical Links Subject to Day-to-Day Disruptions
Rep2: 2018 TranspPol | Transportation resilience to climate change and extreme weather events – Beyond risk and robustness
Rep3: 2015 TRR | Resilience of Coastal Transportation Networks Faced with Extreme Climatic Events

Topic 135 | grade B | fails: F2 | n=169

NPMI -0.448 | Jaccard 0.818 | modularity 0.324
Top-10: signs, sign, retroreflectivity, visibility, marking, markings, guide, wet, lighting, retroreflective
Journal: TRR 0.75 | peak 2003 | slope -0.036pp/yr | nonfixed 0.04
Rep1: 2013 TR-F | Comprehension of traffic signs with symbolic versus text displays
Rep2: 2009 TRR | Effect of Luminance and Text Size on Information Acquisition Time from Traffic Signs
Rep3: 2000 TRR | Minimum In-Service Retroreflectivity of Pavement Markings

Topic 136 | grade A | fails: none | n=303

NPMI -0.114 | Jaccard 0.818 | modularity 0.148
Top-10: pavement, concrete, temperature, pavements, thermal, asphalt, seasonal, moisture, cracking, jointed
Journal: TRR 0.92 | peak 2001 | slope -0.052pp/yr | nonfixed 0.05
Rep1: 2005 TRR | Predicting Asphalt Pavement Temperature with a Three-Dimensional Finite Element Method
Rep2: 2001 TRR | Using Pavement as Solar Collector: Effect on Pavement Temperature and Structural Response
Rep3: 2014 TRR | Prediction of Asphalt Pavement Temperature Profile with Finite Control Volume Method

Topic 137 | grade A | fails: none | n=860

NPMI -0.139 | Jaccard 0.818 | modularity 0.206
Top-10: lane, rural, lanes, freeway, safety, capacity, highways, roundabouts, turn, intersections
Journal: TRR 0.88 | peak 2000 | slope -0.114pp/yr | nonfixed 0.05
Rep1: 2000 TRR | Predicting Operating Speeds on Tangent Sections of Two-Lane Rural Highways
Rep2: 2000 TRR | Evaluation of the Passing Process on Two-Lane Rural Highways
Rep3: 2005 TRR | Speed Factors on Two-Lane Rural Highways in Free-Flow Conditions

Topic 138 | grade A | fails: none | n=862

NPMI -0.077 | Jaccard 0.818 | modularity 0.072
Top-10: driving, drivers, young, risky, self, anger, older, reported, aggressive, driver
Journal: TR-F 0.47 | peak 2013 | slope -0.007pp/yr | nonfixed 0.03
Rep1: 2001 AAP | Are aggressive people aggressive drivers? A study of the relationship between self-reported general aggressive
Rep2: 2016 AAP | The role of personality traits and driving experience in self-reported risky driving behaviors and accident ri
Rep3: 2014 AAP | Personality and attitudes as predictors of risky driving among older drivers

Topic 139 | grade A | fails: none | n=560

NPMI -0.182 | Jaccard 0.667 | modularity 0.072
Top-10: detection, recognition, video, vision, sign, classification, traffic, deep, videos, surveillance
Journal: T-ITS 0.52 | peak 2023 | slope +0.038pp/yr | nonfixed 0.17
Rep1: 2022 IJTST | Indian traffic sign detection and recognition using deep learning
Rep2: 2018 TRR | Traffic Congestion Detection from Camera Images using Deep Convolution Neural Networks
Rep3: 2007 T-ITS | Road-Sign Detection and Recognition Based on Support Vector Machines

Topic 140 | grade A | fails: none | n=636

NPMI -0.144 | Jaccard 0.667 | modularity 0.113
Top-10: transport, projects, infrastructure, cost, public, benefit, freight, rail, european, financing
Journal: TranspPol 0.20 | peak 2000 | slope -0.060pp/yr | nonfixed 0.17
Rep1: 2010 JTG | The broader economic consequences of transport infrastructure investments
Rep2: 2016 ETRR | How to make modal shift from road to rail possible in the European transport market, as aspired to in the EU T
Rep3: 2001 TranspRev | Economic appraisal of European transport projects: the state-of-the-art revisited

Topic 141 | grade A | fails: none | n=619

NPMI -0.029 | Jaccard 0.818 | modularity 0.145
Top-10: control, traffic, ramp, metering, signal, freeway, perimeter, variable, coordinated, predictive
Journal: TR-C 0.30 | peak 2016 | slope +0.006pp/yr | nonfixed 0.05
Rep1: 2010 TransSci | Optimal Motorway Traffic Flow Control Involving Variable Speed Limits and Ramp Metering
Rep2: 2005 TR-C | Model predictive control for optimal coordination of ramp metering and variable speed limits
Rep3: 2011 TRR | Novel Freeway Traffic Control with Variable Speed Limit and Coordinated Ramp Metering

Topic 142 | grade A | fails: none | n=500

NPMI 0.014 | Jaccard 0.818 | modularity 0.083
Top-10: emissions, fuel, duty, diesel, consumption, emission, heavy, gas, light, gasoline
Journal: TR-D 0.42 | peak 2009 | slope -0.029pp/yr | nonfixed 0.12
Rep1: 2004 TRR | Modal Emissions Model for Heavy-Duty Diesel Vehicles
Rep2: 2017 TR-F | Reducing fuel consumption and carbon emissions through eco-drive training
Rep3: 2006 TRR | Comparison of Real-World Fuel Use and Emissions for Dump Trucks Fueled with B20 Biodiesel Versus Petroleum Die

Topic 143 | grade A | fails: none | n=331

NPMI -0.100 | Jaccard 0.818 | modularity 0.155
Top-10: household, travel, trip, demand, forecasting, activity, transferability, survey, generation, models
Journal: TRR 0.93 | peak 2000 | slope -0.087pp/yr | nonfixed 0.01
Rep1: 2004 TRR | Sequential Logit Dynamic Travel Demand Model for Hurricane Evacuation
Rep2: 2013 TRR | Transferability of Freight Trip Generation Models
Rep3: 2003 TRR | Impact of Underreporting on Mileage and Travel Time Estimates: Results from Global Positioning System-Enhanced

Topic 144 | grade B | fails: F2 | n=597

NPMI -0.379 | Jaccard 0.818 | modularity 0.118
Top-10: cycling, cyclists, bicycle, bicycling, walking, bicyclists, cyclist, school, gender, cycle
Journal: TR-F 0.18 | peak 2019 | slope +0.031pp/yr | nonfixed 0.15
Rep1: 2022 TranspPol | More than walking and cycling: What is ‘active travel’?
Rep2: 2008 TRR | Cycling for Everyone
Rep3: 2014 IJST | Cycling Under Influence: Summarizing the Influence of Perceptions, Attitudes, Habits, and Social Environments

Topic 145 | grade A | fails: none | n=900

NPMI -0.043 | Jaccard 0.818 | modularity 0.103
Top-10: pedestrian, crossing, pedestrians, crosswalks, street, crossings, signalized, intersections, safety, behavior
Journal: TRR 0.33 | peak 2016 | slope +0.005pp/yr | nonfixed 0.09
Rep1: 2018 TRR | Pedestrian Behavior at Signalized Intersection Crosswalks: Observational Study of Factors Associated with Dist
Rep2: 2011 AAP | The link between built environment, pedestrian activity and pedestrian–vehicle collision occurrence at signali
Rep3: 2014 JTTE | Study on pedestrian crossing behavior at signalized intersections

Topic 146 | grade A | fails: none | n=281

NPMI -0.111 | Jaccard 0.667 | modularity 0.131
Top-10: concrete, reinforced, fiber, polymer, bridge, strength, cement, decks, curing, beams
Journal: TRR 0.97 | peak 2000 | slope -0.068pp/yr | nonfixed 0.02
Rep1: 2012 TRR | Performance of Steel Fiber–Reinforced Concrete Pipes
Rep2: 2002 TRR | Concrete Bridge Decks Constructed with Fiber-Reinforced Polymer Stay-in-Place Forms and Grid Reinforcing
Rep3: 2006 TRR | Cost-Effective, Structural Stay-in-Place Formwork System of Fiber-Reinforced Polymer for Accelerated and Durab

Topic 147 | grade C | fails: F1,F2 | n=305

NPMI -0.604 | Jaccard 0.429 | modularity 0.228
Top-10: transit, transportation, paratransit, california, north, traveler, successful, american, information, tribal
Journal: TRR 0.84 | peak 2000 | slope -0.108pp/yr | nonfixed 0.01
Rep1: 2019 TRR | Uber Economics: Evaluating the Monetary and Travel Time Trade-Offs of Transportation Network Companies and Tra
Rep2: 2008 TRR | Transit Use at Transit-Oriented Developments in Portland, Oregon, Area
Rep3: 2003 TRR | Transit Use and Proximity to Rail: Results from Large Employment Sites in the San Francisco, California, Bay A

Topic 148 | grade A | fails: none | n=488

NPMI -0.073 | Jaccard 0.742 | modularity 0.099
Top-10: control, adaptive, tracking, sliding, steering, stability, wheel, actuator, suspension, lateral
Journal: T-ITS 0.42 | peak 2023 | slope +0.046pp/yr | nonfixed 0.52
Rep1: 2016 T-ITS | Robust H_∞ Path Following Control for Autonomous Ground Vehicles With Delay and Data Dropout
Rep2: 2022 T-ITS | Fixed-Time Prescribed Performance Adaptive Fixed-Time Sliding Mode Control for Vehicular Platoons With Actuato
Rep3: 2022 T-TE | A Path Following Lateral Control Scheme for Four-Wheel Independent Drive Autonomous Vehicle Using Sliding Mode

Topic 149 | grade A | fails: none | n=242

NPMI -0.007 | Jaccard 0.818 | modularity 0.169
Top-10: lidar, localization, visual, inertial, mapping, odometry, map, monocular, slam, camera
Journal: T-ITS 0.54 | peak 2025 | slope +0.023pp/yr | nonfixed 0.38
Rep1: 2022 T-ITS | Automatic Extrinsic Calibration Method for LiDAR and Camera Sensor Setups
Rep2: 2021 T-ITS | A Comparative Analysis of LiDAR SLAM-Based Indoor Navigation for Autonomous Vehicles
Rep3: 2017 T-IV | Simultaneous Localization and Mapping: A Survey of Current Trends in Autonomous Driving